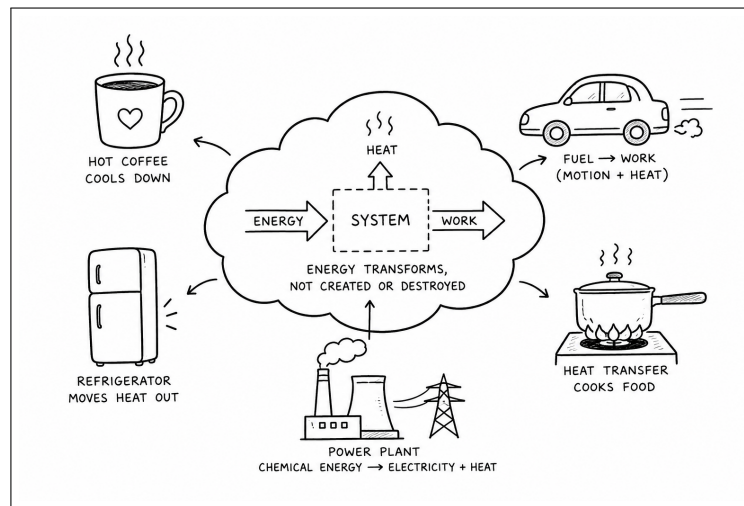




CHAPTER 1 : INTRODUCTION & BASIC CONCEPTS

This chapter introduces the fundamental concepts and vocabulary of thermodynamics needed for understanding energy and its transformations. It develops the basic ideas of systems, control volumes, properties, equilibrium, processes, and thermodynamic states. The chapter also establishes essential foundations involving dimensions, units, temperature, pressure, density, and specific properties. Together, these concepts provide the framework required for analyzing thermodynamic systems and solving engineering problems systematically.

CHAPTER TREE

- | — 1-1 ■ THERMODYNAMICS AND ENERGY
- | — 1-2 ■ IMPORTANCE OF DIMENSIONS AND UNITS
- | — 1-3 ■ SYSTEMS AND CONTROL VOLUMES
- | — 1-4 ■ PROPERTIES OF A SYSTEM
- | — 1-5 ■ DENSITY AND SPECIFIC GRAVITY
- | — 1-6 ■ STATE AND EQUILIBRIUM
- | — 1-7 ■ PROCESSES AND CYCLES
- | — 1-8 ■ TEMPERATURE AND THE ZEROth LAW OF THERMODYNAMICS
- | — 1-9 ■ PRESSURE
- | — 1-10 ■ PRESSURE MEASUREMENT DEVICES
- | — 1-11 ■ PROBLEM-SOLVING TECHNIQUE

1.1 THERMODYNAMICS AND ENERGY

Thermodynamics is the science concerned with energy, energy transformations, and the interaction between energy and matter.

THERMODYNAMICS

Energy may be viewed as the **ability to cause changes**.

The word thermodynamics originates from the Greek words therme (heat) and dynamis (power), reflecting the early development of the subject around converting heat into power.

Today, thermodynamics has a much broader scope and deals with:

- Energy and its different forms.
- Energy transformations.
- Power generation.
- Refrigeration.
- Relationships among the properties of matter.

CONSERVATION OF ENERGY

The **conservation of energy principle** states that during an interaction, energy can change from one form to another, but the total amount of energy remains constant.

Energy cannot be created or destroyed; it can only change forms.

For any system, the change in its energy content is equal to the difference between the energy entering and leaving the system.

$$\Delta E = E_{\text{in}} - E_{\text{out}}$$

Example

When a rock falls from a height, its potential energy is converted into kinetic energy. The total energy is conserved even though the form of energy changes.

FIRST AND SECOND LAWS OF THERMODYNAMICS

The two fundamental laws of thermodynamics describe important characteristics of energy.

Law	Main principle
First law	Energy is conserved.
Second law	Energy has both quantity and quality.

The

first law of thermodynamics is an expression of the conservation of energy principle and establishes energy as a thermodynamic property.

The

second law of thermodynamics states that energy has quality as well as quantity. Actual processes occur in the direction of decreasing energy quality.

Example

A hot cup of coffee placed in a cooler environment eventually loses heat and becomes cooler. However, a cool cup of coffee in the same environment does not spontaneously become hot.

Thus, energy naturally tends to be transferred from a region of higher temperature to a region of lower temperature, resulting in degradation of the usefulness or quality of energy.

HISTORICAL DEVELOPMENT

Although thermodynamic principles have always existed in nature, thermodynamics emerged as a scientific discipline with the development of atmospheric steam engines.

- Thomas Savery constructed a successful atmospheric steam engine in 1697.
- Thomas Newcomen developed another atmospheric steam engine in 1712.
- These early engines were slow and inefficient, but they initiated the development of thermodynamics as a science.
- The first and second laws emerged during the 1850s, primarily through the work of William Rankine, Rudolph Clausius, and Lord Kelvin.
- The term thermodynamics was first used in a publication by Lord Kelvin in 1849.
- The first thermodynamics textbook was written by William Rankine in 1859.

MACROSCOPIC AND MICROSCOPIC APPROACHES

Matter consists of a very large number of particles called molecules. The properties observed at the macroscopic level arise from the behavior of these particles.

For example, the pressure of a gas results from momentum transfer between gas molecules and the walls of its container.

Two approaches can be used to study thermodynamics:

Approach	Basis	Characteristic
Classical thermodynamics	Macroscopic behavior	Direct and relatively simple
Statistical thermodynamics	Average microscopic behavior	More involved

Classical thermodynamics

Classical thermodynamics does not require knowledge of the behavior of individual particles. It studies measurable macroscopic quantities and provides a direct and convenient method for solving engineering problems.

Statistical thermodynamics

Statistical thermodynamics is based on the average behavior of large groups of individual particles. It is a microscopic approach and is more involved than classical thermodynamics. In engineering applications, classical thermodynamics is generally the primary approach, while statistical thermodynamics may be used in a supporting role.

APPLICATION AREAS OF THERMODYNAMICS

Interactions between energy and matter occur throughout nature. Consequently, thermodynamics is relevant to a wide range of engineering systems and everyday phenomena.

Area	Examples
Human body	Metabolic energy conversion and heat rejection
Household systems	Heating, air conditioning, refrigerators, water heaters
Appliances	Pressure cookers, irons, computers, televisions
Transportation	Automobile engines, aircraft, rockets, jet engines
Energy systems	Conventional and nuclear power plants, solar collectors
Industrial systems	Pumps, piping networks, and other thermal systems

Human Body

The human body continuously converts energy within its cells and rejects heat to the surroundings.

Human comfort depends strongly on the rate at which metabolic heat is rejected. Clothing can be adjusted according to environmental conditions to help control this heat transfer.

Household Applications

Thermodynamic principles are involved in the design and operation of many household devices and systems, including:

- Electric and gas ranges.
- Heating and air-conditioning systems.
- Refrigerators.
- Humidifiers.
- Pressure cookers.
- Water heaters.
- Showers.

- Irons.
- Computers and televisions.

Engineering Applications

Thermodynamics plays an important role in the design and analysis of:

- Automotive engines.
- Rockets and jet engines.
- Conventional and nuclear power plants.
- Solar collectors.
- Cars and aircraft.

For example, an energy-efficient house can be designed by minimizing heat loss during winter and heat gain during summer.

The design of computer cooling systems also involves thermodynamic analysis. The size, location, and power input of a cooling fan can be selected based on such analysis.

KEY IDEAS

Thermodynamics : Thermodynamics is the science of energy, its transformations, and its interaction with matter.

Conservation of Energy : Energy cannot be created or destroyed. It can only be transferred or transformed from one form into another.

First Law : The first law expresses conservation of energy and establishes energy as a thermodynamic property.

Second Law : The second law accounts for the quality of energy and indicates that actual processes proceed toward decreasing energy quality.

Classical Thermodynamics : The macroscopic approach does not require knowledge of individual particles and provides a direct method for engineering analysis.

Statistical Thermodynamics : The microscopic approach studies the average behavior of large groups of particles and is more involved.

Applications : Thermodynamics is fundamental to systems involving energy and matter, including human bodies, household appliances, vehicles, power plants, and industrial systems.